

HYG Database Analysis

In this project I analyse the HYG Database of stars to answer the following questions:

- Are the stars which are visible from earth evenly distributed between the hours of right ascension?
- Are the stars which are visible from earth evenly distributed between the northern and southern celestial hemispheres?
- In which hour of right ascension, and in which hemisphere, are there the most visible stars?

I do this in the hope that it will give me a deeper understanding of the positions of the stars around us and the times at which they can be viewed.

The Dataset

The HYG Database (v4.1) is a compilation of the Hipparcos, Yale Bright Star, and Gliese catalogues, and holds extensive data of over 119,000 stars (with 37 columns for various IDs, names, and information regarding position, brightness, and motion, among others). Only a subset are relevant to my research, just 7 columns by 8920 rows (these are the stars with a magnitude of 6.5 or below, so are visible by eye from earth).

I acquired the original data from: <https://www.astronexus.com/projects/hyg>.

Below I have presented a brief artistic overview of the data through 3 graphs.

One of the issues with this database is that not all distances are collected or certain, and for these datapoints the distance is recorded as 100,000. So below I have found that roughly 2.3% of the data has to be omitted from the first visualisation.

```
In [1]: ## imports
import gzip
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import matplotlib.patches as mpatches
import textwrap
from matplotlib.patches import Patch
from matplotlib.patches import Rectangle
from PIL import Image
from scipy.stats import ks_2samp
```

```
In [2]: ## reading CSV and basic filtering
with gzip.open("C:/Users/Sarah/OneDrive/Documents/Coding/Star polar analysis/hygdat
og_data=pd.read_csv(f)
```

```
slim_data= og_data[[ 'proper', 'con', 'ra', 'dec', 'mag', 'absmag','dist', 'spect']]
bri_data = slim_data[slim_data["mag"] <= 6.5].sort_values(by="mag")
```

```
In [3]: ## cleaning inaccurate distance records
graph_data= bri_data[bri_data["dist"] < 100000].copy()
print(f"\nTotal number of stars: {len(bri_data)}"
      f"\nNumber of stars with recorded distance data: {len(graph_data)}"
      f" ({round(len(graph_data) / len(bri_data) *100, 1)}%\n")
```

Total number of stars: 8921

Number of stars with recorded distance data: 8715 (97.7%)

```
In [4]: ## graph 1
theta2 = np.deg2rad(graph_data["ra"].values *16)
radii2 = graph_data["dist"].values
color_map = {
    "O": "#5A7ADB", "B": "#89CFF0", "A": "#F8F8F8", "F": "#FFD700", "G": "#FFAA
graph_data["color"] = graph_data["spect"].astype(str).apply(lambda x: color_map.get
size_map = {
    "Ia": 8, "Ib": 7, "II": 6, "III": 5, "IV": 4, "V": 3, "D": 2 }

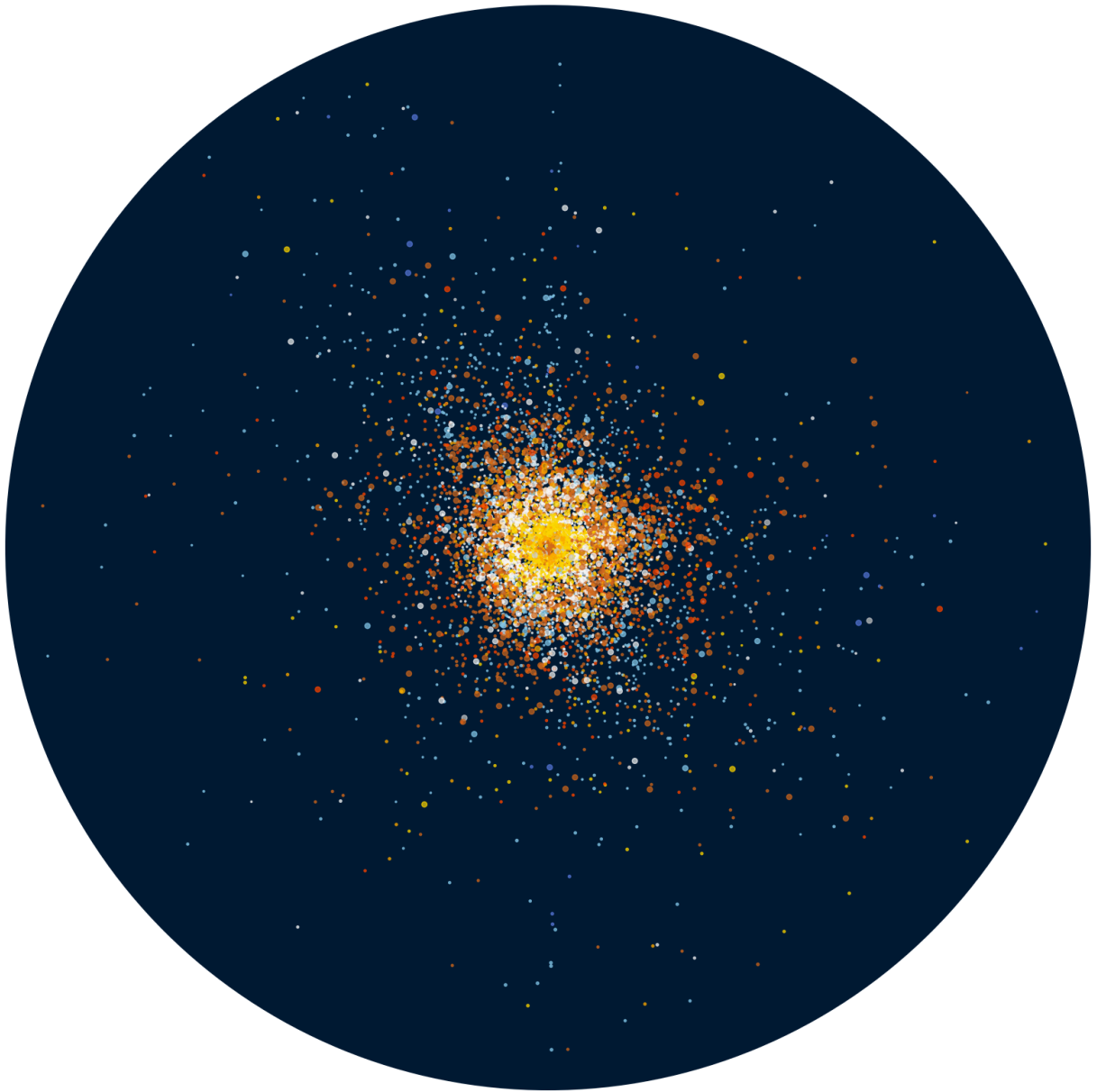
def get_size(spect):
    for key in size_map:
        if key in spect:
            return size_map[key]
    return 30

graph_data["size"] = graph_data["spect"].astype(str).apply(get_size)

plt.figure(figsize=(20, 20))
ax = plt.subplot(projection='polar')
ax.scatter(theta2, radii2, color=graph_data["color"], s=graph_data["size"], edgecol
x_labels= ['0', '3', '6', '9', '12', '15', '18', '21']
bottom=300

plt.title("Visible Stars of the HYG Database", fontsize=40, pad=70)
ax.grid(False)
ax.set_xticklabels([])
ax.set_yticklabels([])
ax.spines['polar'].set_visible(False)
ax.set_facecolor("#011a35")
plt.show()
plt.clf()
```

Visible Stars of the HYG Database



<Figure size 640x480 with 0 Axes>

```
In [5]: ## dropping the sun  
bri_data = bri_data.drop(bri_data[bri_data["mag"] == -26.70].index)
```

```
In [6]: ## graph 2: aggregating stats, defining split lists, and splitting lists  
con_data_c=bri_data.groupby("con").agg(  
    ra_range=("ra", lambda x: round(x.max() - x.min(), 2)),  
    ra_min=("ra", lambda x: round(x.min(), 2)),  
    ra_max=("ra", lambda x: round(x.max(), 2)),  
    dec_avg=("dec", lambda x: round(x.mean(), 2)),  
    dec_min=("dec", lambda x: round(x.min(), 2)),  
    dec_max=("dec", lambda x: round(x.max(), 2)),  
    dec_range=("dec", lambda x: round(x.max() - x.min(), 2)),  
    ra_data=("ra", lambda x: sorted(list(map(lambda y: round(y, 2), x)))),  
    dist_avg=("dist", lambda x: round(x.mean(), 2))
```

```

)
con_data_c=con_data_c.reset_index()
crossing_con = con_data_c[con_data_c["ra_max"] - con_data_c["ra_min"] > 22].copy()

def split_list(lst):
    if len(lst) < 2:
        return lst, []
    diffs = [lst[i+1] - lst[i] for i in range(len(lst)-1)]
    split_index = diffs.index(max(diffs)) + 1
    return lst[:split_index], lst[split_index:]
crossing_con[['early', 'late']] = crossing_con['ra_data'].apply(lambda x: pd.Series

```

```

In [7]: ## graph 2: adjusting mins and maxes for half boxes, merging dfs
earlies = crossing_con.copy()
lates = crossing_con.copy()

earlies["ra_min"]= 0
earlies["ra_max"] = earlies["early"].apply(lambda x: x[-1])
earlies["ra_range"]= earlies["ra_max"]

lates["ra_min"]= lates["late"].apply(lambda x: x[0])
lates["ra_max"] = 23.99
lates["ra_range"]= lates["ra_max"] - lates["ra_min"]

nolist=crossing_con["con"].tolist()
con_data_d=con_data_c[~con_data_c["con"].isin(nolist)]
con_data_e = pd.concat([con_data_d, earlies, lates], ignore_index=True)
con_graph_data= con_data_e.loc[:, ["con", "ra_min", "ra_range", "ra_max", "dec_min"]

```

```

In [8]: ## graph 2: plotting constellations
fig, ax = plt.subplots(figsize=[9,7])
chosen_cons=["Ari", "Tau", "Gem", "Cnc", "Leo", "Vir", "Lib", "Sco", "Sgr", "Cap", "
con_data_e["facecolor"]= con_data_e["con"].apply(lambda x: "#FFC200" if x == "Psc"
con_data_e["zorder"]= con_data_e["dist_avg"].apply(lambda x: 4 if x<100 else 3 if x
for _, row in con_data_e.iterrows():
    rect = Rectangle((row["ra_min"], row["dec_min"]), row["ra_range"], height=row["
        alpha=0.7, zorder=row["zorder"])
    ax.add_patch(rect)

ax.set_xlabel("Right Ascension (hours)", labelpad=10)
ax.set_ylabel("Declination (degrees)", labelpad=10)
custom_ticks = [0, 3, 6, 9, 12, 15, 18, 21, 24]
plt.xticks(custom_ticks)
ax.set_xlim(0, 24)
ax.set_ylim(-95, 95)
ax.set_title("The Path of the Sun Through the Constellations", pad=20)
ax.axhline(y=0, linestyle="--", color="#011a35", alpha=0.7, zorder=10)

fig.subplots_adjust(right=0.75) # Adjust layout to fit text box

box_1 = textwrap.fill( "Throughout the year as the Earth moves, the Sun appears t
                        width=30 )

box_2 = textwrap.fill( "Look up at the stars at night on spring equinox (hour 0).
                        width=30 )

```

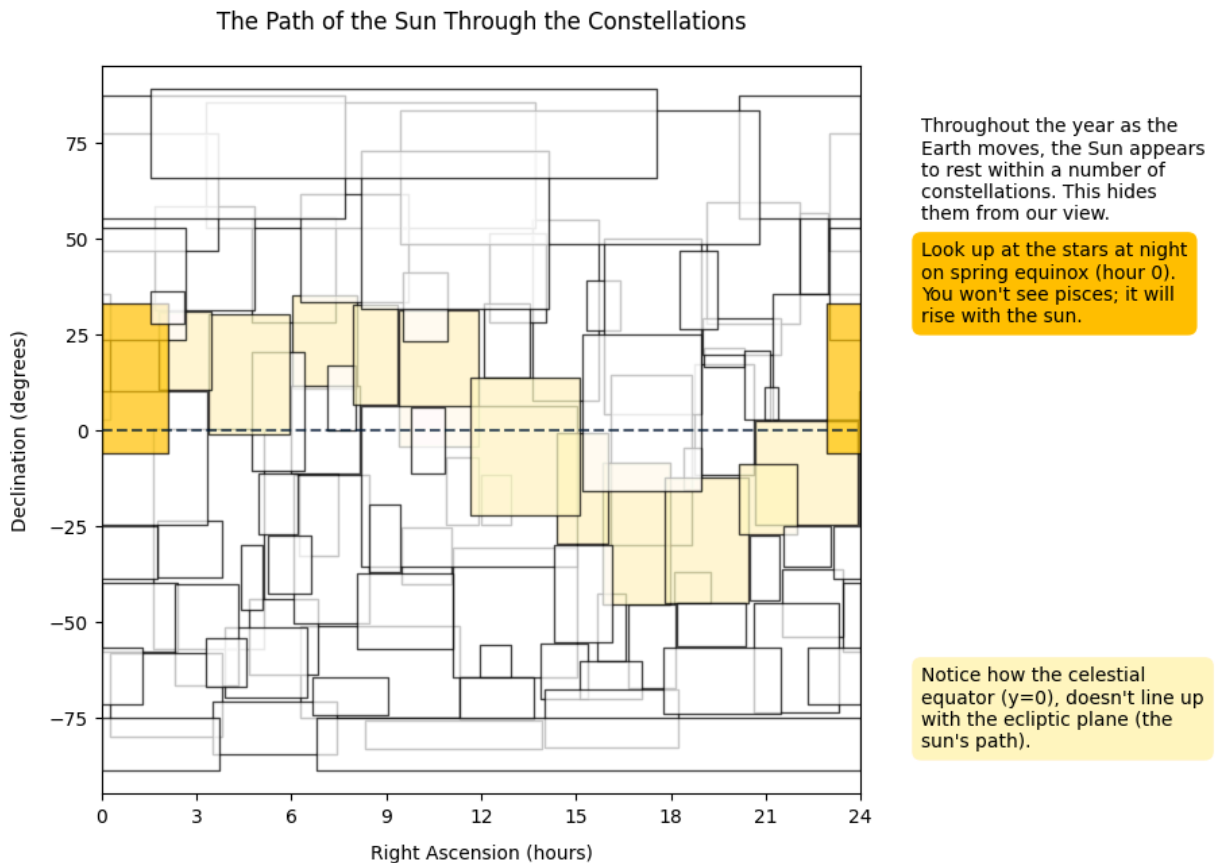
```

box_3 = textwrap.fill( """Notice how the celestial equator (y=0), doesn't line up w
width=30)

fig.text( 0.8, 0.77, box_1, fontsize=10, verticalalignment='center', horizontalalign
        bbox=dict(facecolor='white', edgecolor='none', boxstyle='round,pad=0.5'))
fig.text( 0.8, 0.65, box_2, fontsize=10, verticalalignment='center', horizontalalign
        bbox=dict(facecolor='#FFC200', edgecolor='none', boxstyle='round,pad=0.5'))
fig.text( 0.8, 0.2, box_3, fontsize=10, verticalalignment='center', horizontalalign
        bbox=dict(facecolor="#FFF7C0", edgecolor='none', boxstyle='round,pad=0.5'))

plt.show()

```



```

In [9]: ## graph 3
dist_mag_data= graph_data.loc[:, ["proper", "con", "ra", "dec", "mag", "dist", "abs
dist_mag_data = dist_mag_data.drop(dist_mag_data[dist_mag_data["mag"] == -26.70].in

legend_handles = [mpatches.Patch(color=color, label=spec_type) for spec_type, color

fig, ax=plt.subplots(figsize=[9,7])
ax.scatter(dist_mag_data["dist"], dist_mag_data["absmag"], s=2, color=dist_mag_data

box_1 = textwrap.fill( """This dataset is defined by magnitude, as only visible sta
width=30 )

box_2 = textwrap.fill( """This means the results show the Malmquist Bias: """,
width=30 )

box_3 = textwrap.fill( """Unlike the actual distribution, the closer stars in the d

```

```

width=30 )

box_4 = textwrap.fill( """This results in this curve and clear groupings of A, F, a
width=30 )

fig.text( 0.95, 0.77, box_1, fontsize=10, verticalalignment='center', horizontalali
bbox=dict(facecolor='white', edgecolor='none', boxstyle='round,pad=0.5'))

fig.text( 0.95, 0.68, box_2, fontsize=10, verticalalignment='center', horizontalali
bbox=dict(facecolor="white", edgecolor='#011a35', boxstyle='round,pad=0.5')

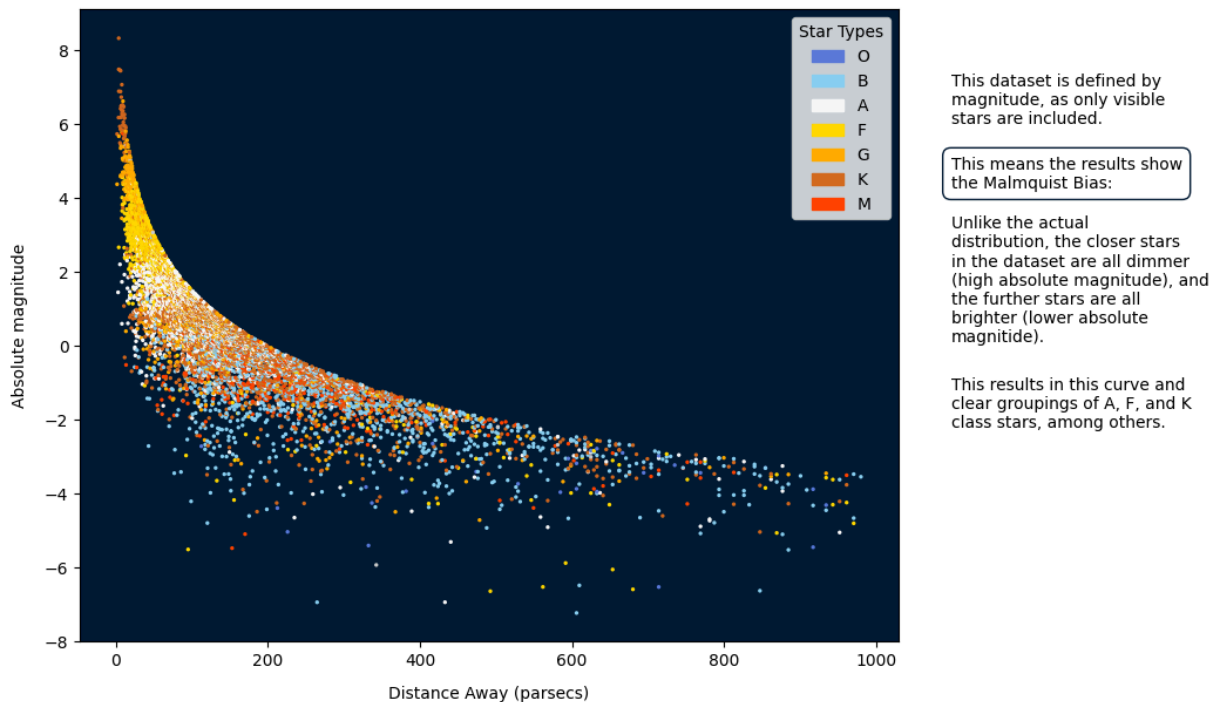
fig.text( 0.95, 0.55, box_3, fontsize=10, verticalalignment='center', horizontalali
bbox=dict(facecolor='white', edgecolor='none', boxstyle='round,pad=0.5'))

fig.text( 0.95, 0.4, box_4, fontsize=10, verticalalignment='center', horizontalali
bbox=dict(facecolor='white', edgecolor='none', boxstyle='round,pad=0.5'))

ax.set_facecolor("#011a35")
ax.set_ylabel("Absolute magnitude", labelpad=10)
ax.set_xlabel("Distance Away (parsecs)", labelpad=10)
ax.set_title("The Effect of Distance on Visibility", pad=20, size=14)
ax.legend(legend_handles, color_map, loc="upper right", title="Star Types")
plt.show()

```

The Effect of Distance on Visibility

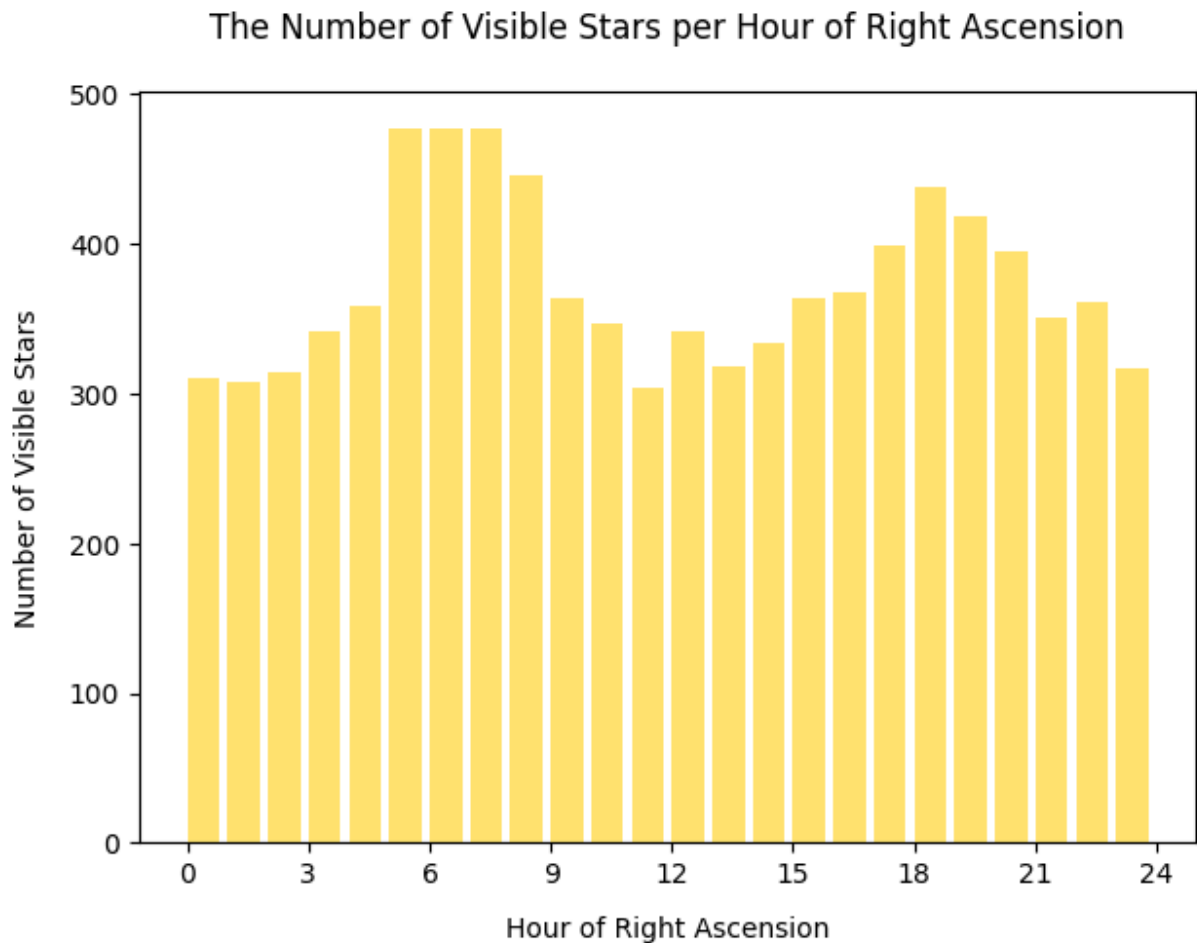


Question 1: distrubution by right ascension

Below I plot the distribution of visible stars by hour of right ascension, first as a bar chart, and second as a polar for more concept clarity.

```
In [10]: ## grouping to find star counts
bri_data["RA hour"] = bri_data["ra"].astype(int)
bri_counts=bri_data.groupby("RA hour").agg(count=("RA hour","count")).reset_index()
```

```
In [11]: ## graph 4
fig, ax=plt.subplots(figsize=[7,5])
custom_ticks = [0, 3, 6, 9, 12, 15, 18, 21, 24]
ax.bar(bri_counts["RA hour"], bri_counts["count"], color="#FFE470", align='edge')
ax.set_xticks(custom_ticks)
ax.set_xlabel("Hour of Right Ascension", labelpad=10, fontsize=10)
ax.set_ylabel("Number of Visible Stars", labelpad=10, fontsize=10)
ax.set_title("The Number of Visible Stars per Hour of Right Ascension", pad=20)
ax.grid(False, which="major")
```



```
In [12]: ## graph 5
y = bri_counts["count"]
theta=np.linspace(0, 2 * np.pi, num=24, endpoint=False)
radii = y * 1.4

colors= []
for value in y:
    if value <=350 :
        colors.append('#FFF7C0')
    elif value <= 400 :
        colors.append('#FFE470')
    elif value <= 450 :
```

```

colors.append('#FFD000')
else : colors.append('#FFC200')

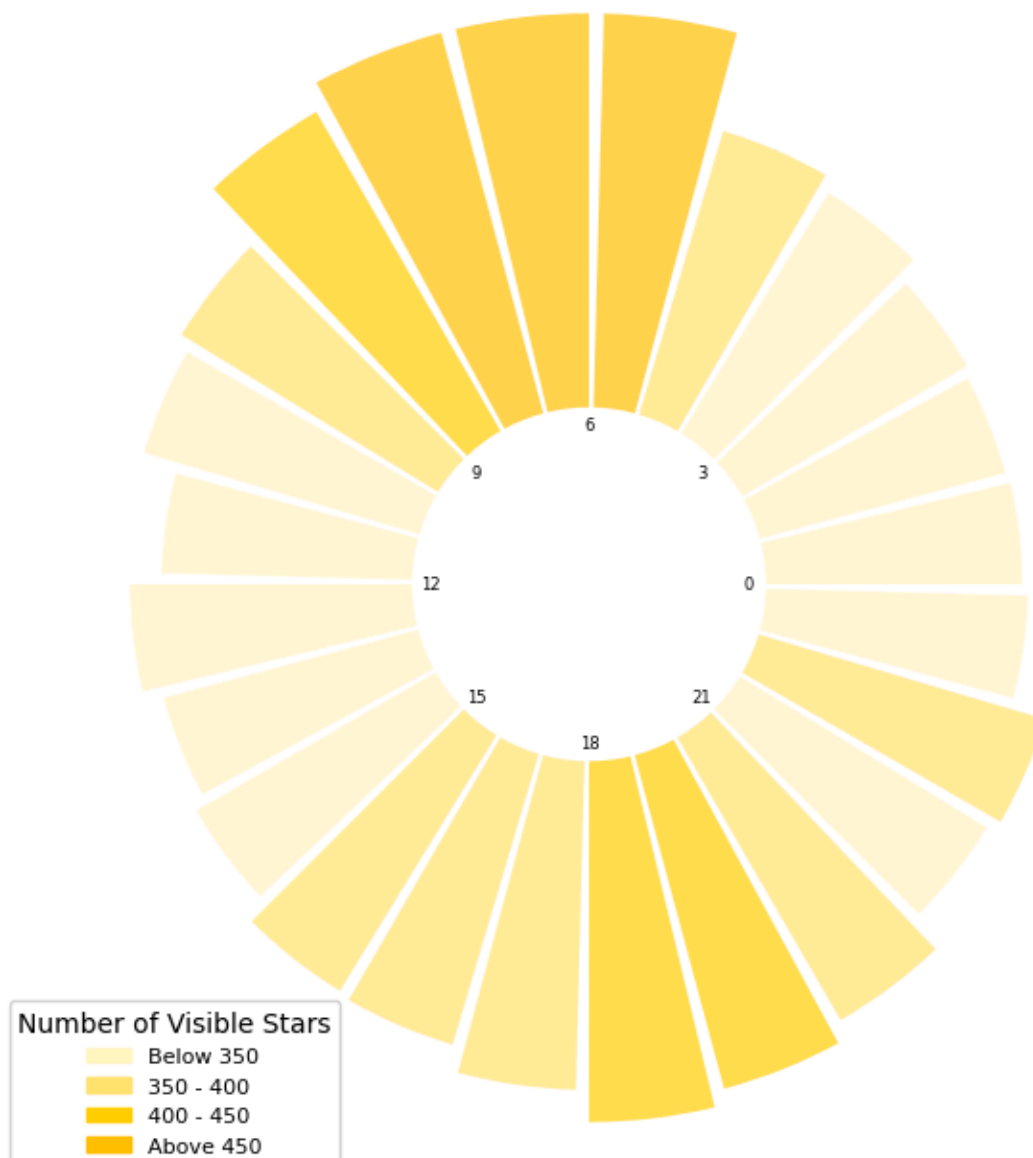
plt.figure(figsize=(8, 8))
ax = plt.subplot(projection='polar')
ax.bar(theta, radii, width=(2 * np.pi/24)*0.9, bottom = 300, color=colors, edgecolor=
x_labels= ['0', ' ', ' ', ' ', '3', ' ', ' ', ' ', '6', ' ', ' ', ' ', '9', ' ', ' ', ' ', 
bottom=300
for i, angle in enumerate(theta):
    r_label= bottom - 30
    ax.annotate(x_labels[i], xy=(angle, r_label), ha='center', va='center', fontsize=14)

leg_handles= [
    Patch( color= "#FFF7C0", label= "Below 350"),
    Patch( color= "#FFE470", label= "350 - 400"),
    Patch( color= "#FFD000", label= "400 - 450"),
    Patch( color= "#FFC200", label= "Above 450")]

ax.legend(loc='lower left', title= "Number of Visible Stars", handles=leg_handles,
plt.title("The Number of Visible Stars per Hour of Right Ascension", fontsize=14, p
ax.grid(False)
ax.set_xticklabels([])
ax.set_yticklabels([])
ax.spines['polar'].set_visible(False)

plt.show()
plt.clf()
```


The Number of Visible Stars per Hour of Right Ascension



<Figure size 640x480 with 0 Axes>

Answer:

It appears that yes, the distribution of visible stars varies between the hours, with peaks in hours 5-8 and 17-20.

Question 2: distribution by declination

I will now refine the results to consider declination. First, the sum of stars in each hemisphere overall, and second, their distribution.

```
In [13]: ## grouping by hemisphere  
bri_data["hemisphere"] = bri_data["dec"] > 0
```

```

bri_data["hemisphere"] = bri_data["hemisphere"].map({True: "Northern", False: "South
dec_counts=bri_data.groupby(["RA hour", "hemisphere"]).agg(count=("RA hour", "count
pos_dec=dec_counts[dec_counts["hemisphere"]=="Northern"]
neg_dec=dec_counts[dec_counts["hemisphere"]=="Southern"]
pos_sum= pos_dec['count'].values.sum()
neg_sum= neg_dec['count'].values.sum()
print(f"\nThe number of visible stars in the northern celestial hemisphere: {pos_su
      f"The number of visible stars in the southern celestial hemisphere: {neg_sum}

```

The number of visible stars in the northern celestial hemisphere: 4275 (47%)

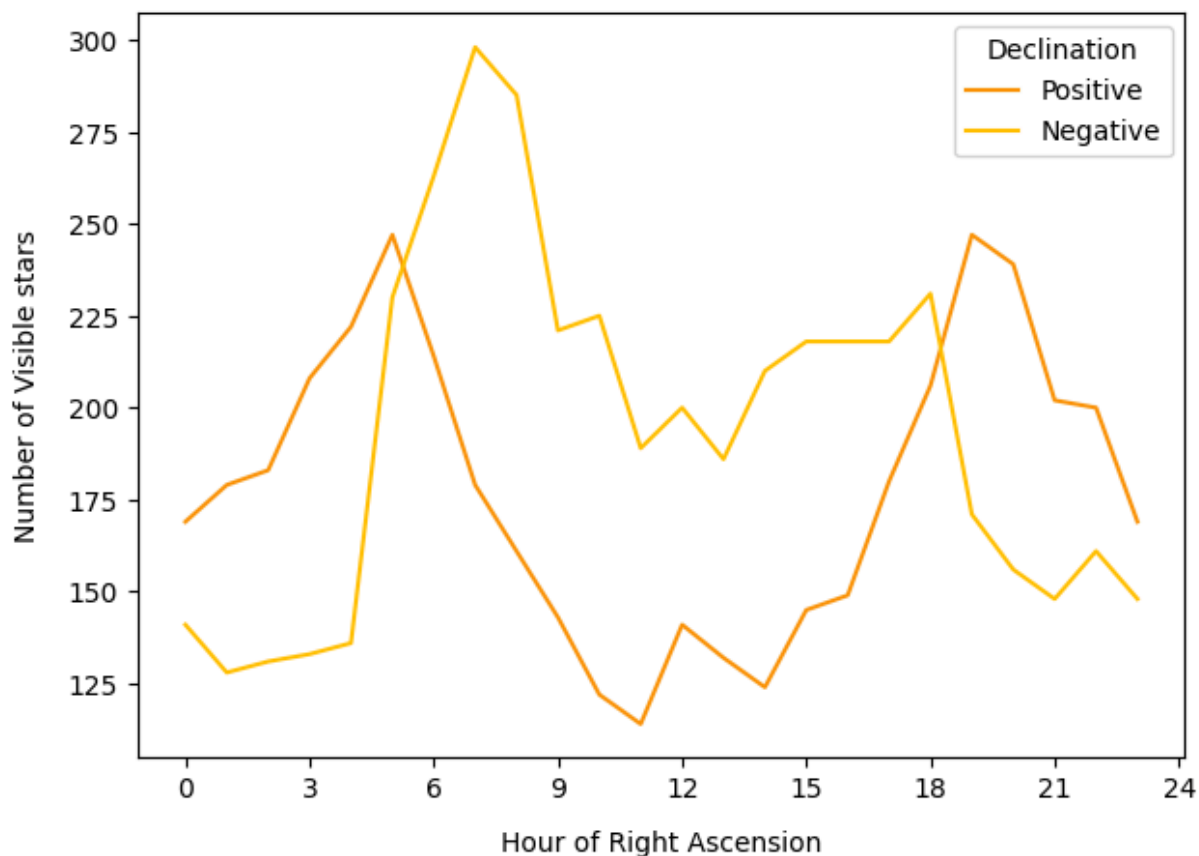
The number of visible stars in the southern celestial hemisphere: 4645 (52%)

```

In [14]: ## graph 6
pos_y = pos_dec["count"].values
neg_y = neg_dec["count"].values
x= pos_dec["RA hour"].values
fig, ax=plt.subplots(figsize=[7,5])
pos_handle, = ax.plot(x, pos_y, label="Positive", color="#fd9509")
neg_handle, = ax.plot(x, neg_y, label="Negative", color="#FFC200")
custom_ticks = [0, 3, 6, 9, 12, 15, 18, 21, 24]
plt.xticks(custom_ticks)
plt.title("The Number of Visible Stars by Right Ascension and Declination", pad=20)
plt.legend(loc='upper right', title= "Declination", handles=[pos_handle, neg_handle]
plt.xlabel("Hour of Right Ascension", labelpad=10)
plt.ylabel("Number of Visible stars", labelpad=10)
plt.show()
plt.clf()

```

The Number of Visible Stars by Right Ascension and Declination



<Figure size 640x480 with 0 Axes>

Answer:

It appears that, yes, the distribution of stars varies between the hemispheres, as the southern has 370 more.

Additionally, while both hemispheres seem to have a larger number of stars in the hours surrounding 6 and 18, the hour bands in between seem to favour one hemisphere over the other. In hours 9-14, there are significantly more stars in the southern, and in 21-4, there are more in the northern.

Question 3: which combination has the most stars

Below I have found the top 5 combinations of RA and Dec to see the most stars:

```
In [15]: ## table 1
ordered_dec_counts = dec_counts.sort_values(by="count", ascending=False).reset_index()
ordered_dec_counts.iloc[:5].style.hide(axis="index")
```

Out[15]:

RA hour	hemisphere	count
7	Southern	298
8	Southern	285
6	Southern	263
5	Northern	247
19	Northern	247

7	Southern	298
8	Southern	285
6	Southern	263
5	Northern	247
19	Northern	247

```
In [16]: ## constellations in hr7 south
neg7= bri_data[(bri_data["hemisphere"]=="Southern") & (bri_data["RA hour"] == 7)]
print(neg7["con"].unique())
```

```
['CMa' 'Pup' 'Car' 'Vol' 'Mon' 'Men']
```

Answer:

The section with the highest number of stars is right scension 7 with a negative declination, which contains stars from the constellations Canis Major, Puppis, Carina, Volans, Monoceros, and Mensa. Interestingly, it seems the top three combinations are in the southern celestial hemisphere, and the top 4 are in neighbouring hours.

Conclusions:

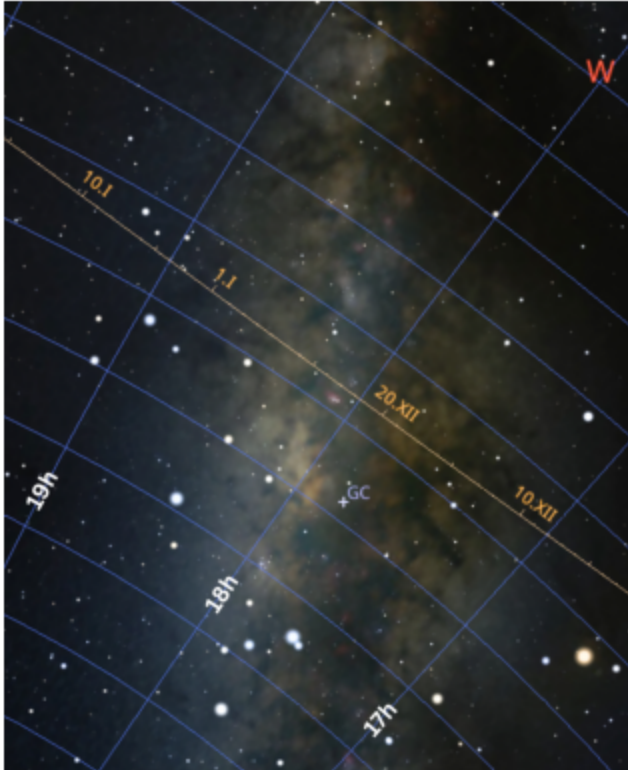
The main questions my data presents to me are these:

- Why does it appear that hours 5-8, and 17-20 have a higher density of stars across both celestial hemispheres?
- Why does the distribution vary between the northern and southern celestial hemispheres?

Now, I am not an astronomer myself, and as such I'm sure there are many wonderful things to learn about the galaxy which go over my head. However, the main understanding I have been able to find it this: these patterns seem to be a result of the position of our solar system within the galaxy, specifically its orientation and the angle of the intersections of the ecliptic and galactic planes.

The intersections between the galactic and celestial planes are in the peak hours I have discovered in my analysis, as you can see in the image from starrarium below:

```
In [17]: plane_inter_1= mpimg.imread("C:/Users/Sarah/OneDrive/Documents/Coding/Star polar an
plt.figure(figsize=(10, 5))
plt.imshow(plane_inter_1)
plt.axis("off")
plt.show()
```



Additionally, as is clear in my findings from goal 2, despite both the northern and southern celestial hemispheres peaking at these hours, they vary in the hours between. In hours 8-17, there are more visible stars in the southern hemisphere, but in hours 20-5 there are more in the northern. This also appears to follow the galactic plane. As you can see below, the milkyway is visible to us in hour 15 in the southern celestial hemisphere (it has a negative declination).

```
In [18]: plane_inter_1= mpimg.imread("C:/Users/Sarah/OneDrive/Documents/Coding/Star polar an
plt.figure(figsize=(10, 5))
plt.imshow(plane_inter_1)
plt.axis("off")
plt.show()
```



The hours 5-9 in particular are especially dense with stars, and I expect this is because it affords us a good view of the Orion arm at the intersection. This arm, like the other arms despite its smaller size, has a 'significant stellar overdensity' (Shen & Zheng, p18), and is considered a stellar nursery, which means it has a large number of young, bright stars (think the Orion Nebular). This gives clear reason for more visible stars in these hours.

This project has given me insight into the times of year and the locations in the sky to see star dense areas, and has also given me a greater knowledge of how our solar system interacts with the wider galaxy, I'm excited to put this understanding into practice when planning travels under the night sky.